Two Samples, One Difference

Topic 5.8 · TODAY'S PROBLEM

Suppose parcel pickup waits have the shown population parameters. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: "South is skewed, so I would not use a normal model for the difference."

Oscar: "North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273."

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE (5 min, on your sheet)

Identify one argument you would retain and one that needs more evidence. Cite a population shape or sample size.

DOK 1 **(a)** State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

DOK 2 **(b)** Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

DOK 3 **(c)** Adjudicate both accounts and decide whether 0.0273 is warranted. Justify each sample mean separately using its shape and sample size, coordinate independence, then use a South sample of 8 to expose the consequence of any rejected shortcut.

Video + follow-along: `u5_lesson8_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.